RF_! OF A FINITE MODULE ALONG A SEPARATED AFFINE SMOOTH RELATIVE CURVE OVER A NOETHERIAN AFFINE BASE HAS CONSTRUCTIBLE COHOMOLOGY SHEAVES (TORSION COEFFICIENTS)

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Good-curves diagram over a generic point (More on Morphisms, Lemma more-morphisms-lemma-make-good-curves, canonical tag 0GK7). cited at proof lines 4775 and 4784 (part (4)). Supplies the auxiliary diagram over the generic point and the dense open W in the fibres around which the whole reduction chain is organised.
- Relative triangle associated to an open (Lemma lemma-relative-triangle-associated-to-open, canonical tag 0GKP). cited at proof lines 4788-4795 and 4807-4813. Gives both distinguished triangles used to discard the quasi-finite complement terms Z and Z'.
- Constructibility for the proper smooth family of curves (Lemma lemma-proper-smooth-family-curves-modules, canonical tag 0GKW). cited at proof lines 4822-4823. The terminal named case to which the induction reduces, combined with finite etale T_i -> V.

Source reference. Original lemma 0GK0: sources/stacks-chow/tags.txt, source lines 20670–20670. Cited at proof line 4745: identifies the stalks of R^qf_!M with the compact cohomology H^q_c of the geometric fibre in the base case d = 0.

Source reference. Original lemma 0GKR: sources/stacks-chow/tags.txt, source lines 20696–20696. Cited at proof line 4746: gives finiteness of H^q_c(X_s, M) as a Lambda-module in the base case.

Source reference. Original lemma 0F7L: sources/stacks-chow/tags.txt, source lines 19011–19011. Cited at proof lines 4753-4754: base change compatibility of Rf_!M lets the induction step be reduced to a dense open U.

Source reference. Original lemma 0GK7: sources/stacks-chow/tags.txt, source lines 20677–20677. Cited at proof lines 4775 and 4784: supplies the auxiliary diagram over the generic point and its part (4) dense open W in the fibres.

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Source reference. Original lemma 095Q: sources/stacks-chow/tags.txt, source lines 11593–11593. Cited at proof lines 4777-4779: since V -> U is finite surjective, constructibility may be checked after pullback to V.

Source reference. Original lemma 0GKP: sources/stacks-chow/tags.txt, source lines 20694–20694. Cited at proof lines 4788-4795 and 4807-4813: the two distinguished triangles that discard the quasi-finite complements Z and Z'.

Source reference. Original lemma 0GK3: sources/stacks-chow/tags.txt, source lines 20673–20673. Cited at proof lines 4796-4798 and 4824-4825: the quasi-finite Z -> V and the finite etale T_i -> V terms have constructible R_!M.

Source reference. Original lemma 0GKQ: sources/stacks-chow/tags.txt, source lines 20695–20695. Cited at proof lines 4801-4804: the thickening nu : nu^{-1}(W) -> W leaves R_!M unchanged.

Source reference. Original lemma 0F7J: sources/stacks-chow/tags.txt, source lines 19009–19009. Cited at proof lines 4816-4821: the second equality factoring R(\overline{Y}_i -> V)_!M as R(T_i -> V)_! composed with R(\overline{Y}_i -> T_i)_!M.

Source reference. Original lemma 0GKW: sources/stacks-chow/tags.txt, source lines 20701–20701. Cited at proof lines 4822-4823: the terminal case, constructibility of R(\overline{Y}_i -> T_i)_!M for the proper smooth family of curves.

Lemma 0.2. *Let S be a Noetherian affine scheme of finite dimension. Let $f : X \to S$ be a separated, affine, smooth morphism of relative dimension 1. Let Λ be a Noetherian ring which is torsion. Let M be a finite Λ-module. Then $Rf_!\underline{M}$ has constructible cohomology sheaves.*

Proof. We will prove the result by induction on $d = \dim(S)$.

Base case. If $d = 0$, then the only thing to show is that the stalks of $R^q f_!\underline{M}$ are finite Λ-modules. If $\overline{s}$ is a geometric point of S, then we have $(R^q f_!\underline{M})_{\overline{s}} = H^q_c(X_{\overline{s}}, \underline{M})$ by Lemma 0GK0. This is a finite Λ-module by Lemma 0GKR.

Induction step. It suffices to find a dense open $U \subset S$ such that $Rf_!\underline{M}|_U$ has constructible cohomology sheaves. Namely, the restriction of $Rf_!\underline{M}$ to the complement $S \setminus U$ will have constructible cohomology sheaves by induction and the fact that formation of $Rf_!\underline{M}$ commutes with all base change (Lemma 0F7L). In fact, let $\eta \in S$ be a generic point of an irreducible component of S. Then it suffices to find an open neighbourhood U of η such that the restriction of $Rf_!\underline{M}$ to U is constructible. This is what we will do in the next paragraph.

Given a generic point $\eta \in S$ we choose a diagram

$$\begin{array}{ccccccccc}
\overline{Y}_1 \amalg \ldots \amalg \overline{Y}_n & \xleftarrow[j]{} & Y_1 \amalg \ldots \amalg Y_n & \xrightarrow[\nu]{} & X_V & \longrightarrow & X_U & \longrightarrow & X \\
 & \searrow & \downarrow & & \downarrow & & \downarrow & & \downarrow f \\
 & & T_1 \amalg \ldots \amalg T_n & \longrightarrow & V & \longrightarrow & U & \longrightarrow & S
\end{array}$$

as in More on Morphisms, Lemma 0GK7. We will show that $Rf_!\underline{M}|_U$ is constructible. First, since $V \to U$ is finite and surjective, it suffices to show that the pullback to V is constructible, see Étale Cohomology, Lemma 095Q. Since formation of $Rf_!$ commutes with base change, we see that it suffices to show that $R(X_V \to V)_!\underline{M}$ is constructible. Let $W \subset X_V$ be the open subscheme given to us

by More on Morphisms, Lemma 0GK7 part (4). Let $Z \subset X_V$ be the reduced induced scheme structure on the complement of W in X_V. Then the fibres of $Z \to V$ have dimension 0 (as W is dense in the fibres) and hence $Z \to V$ is quasi-finite. From the distinguished triangle

$$R(W \to V)_!\underline{M} \to R(X_V \to V)_!\underline{M} \to R(Z \to V)_!\underline{M} \to \ldots$$

of Lemma 0GKP and from Lemma 0GK3 we conclude that it suffices to show that $R(W \to V)_!\underline{M}$ has constructible cohomology sheaves. Next, we have

$$R(W \to V)_!\underline{M} = R(\nu^{-1}(W) \to V)_!\underline{M}$$

because the morphism $\nu : \nu^{-1}(W) \to W$ is a thickening and we may apply Lemma 0GKQ. Next, we let $Z' \subset \coprod \overline{Y}_i$ denote the complement of the open $j(\nu^{-1}(W))$. Again $Z' \to V$ is quasi-finite. Again use the distinguished triangle

$$R(\nu^{-1}(W) \to V)_!\underline{M} \to R(\coprod \overline{Y}_i \to V)_!\underline{M} \to R(Z' \to V)_!\underline{M} \to \ldots$$

to conclude that it suffices to prove

$$R(\coprod \overline{Y}_i \to V)_!\underline{M} = \bigoplus\nolimits_i R(\overline{Y}_i \to V)_!\underline{M} = \bigoplus\nolimits_i R(T_i \to V)_!R(\overline{Y}_i \to T_i)_!\underline{M}$$

has constructible cohomology sheaves (second equality by Lemma 0F7J). The result for $R(\overline{Y}_i \to T_i)_!\underline{M}$ is Lemma 0GKW and we win because $T_i \to V$ is finite étale and we can apply Lemma 0GK3. $\square$